Name: _____

Student ID: _____

CEC 300 Exam 1

Instructions: You may submit this document or create your own document with your solutions. Please add your name and studentID to the top of the document. Answer each question to the best of your abilities. I grade what I can see, so make sure you include all the details you wish to convey. Your submission should be your own. ChatGPT and other AI services are not permitted (with a penalty up to a zero on the exam). Please type written responses. The exam is due via Canvas submission on Monday, March 30 by 11:59pm as a PDF.

Essay Question Grading Rubric:

Each response must be typed. The points for each essay question are divided as follows:

1. **Technical Accuracy (10 points)**
   - The response must be technically correct, demonstrating a clear understanding of the principles, components, and concepts discussed in the prompt.
   - The explanation should be accurate and reflect a deep comprehension of the subject matter.
2. **Completeness (10 points)**
   - The response must address each part of the prompt thoroughly.
   - All aspects of the question should be covered in detail, ensuring no part of the prompt is overlooked.
3. **Citation and References (3 points)**
   - All references must be cited in IEEE format.
   - If course slides are referenced, they should be cited as follows: "CEC 300. 'slide title', URL: http://sourceURL.com, Last Accessed: MM-DD-YYYY.
   - Proper citation demonstrates the ability to integrate and acknowledge sources correctly.
4. **Organization and Clarity (2point)**
   - The response should be well-organized with clear and logical flow.
   - Paragraphs should be coherent, and ideas should be presented in a structured manner.
   - The writing should be clear and concise, making it easy to follow.

Essay 1 [25 points]: Uncrewed Automated Aircraft Navigation

In a "GPS denied-environment" logic test, you are tasked with designing the navigational suite for an autonomous Cessna 182 (i.e., functions of the pilot are replaced with automation). The objective is to achieve full-mission automation—from throttle-up to touchdown—without any access to Global Navigation Satellite Systems (GNSS/GPS). Your system must rely exclusively on the terrestrial National Airspace System (NAS) infrastructure and the aircraft's onboard analog receivers.

The available sensor suite includes:

1. **NDB / ADF** (Non-Directional Beacon / Automatic Direction Finder)
2. **VOR** (VHF Omnidirectional Range)
3. **DME** (Distance Measurement Equipment)
4. **ILS** (Instrument Landing System: Localizer and Glideslope)

The Assignment

Develop a technical proposal identifying which specific NAVAID (or combination thereof) your autonomous logic will prioritize for each of the following five flight phases. You should justify each by describing not only the suitable sensors, but what specific information they can provide to support localization or navigation.

- **Departure & Climb-to-Altitude:** (Focus on transitioning from the terminal area to the airway).
- **En Route (Route Following):** (Focus on long-distance station-to-station tracking).
- **Airport Arrival / Terminal Area:** (Focus on the transition from high-altitude airways to the approach environment).
- **Final Approach & Landing:** (Focus on high-precision vertical and lateral guidance).

Demonstrate your understanding of these technologies and how they can be applied to the aircraft’s operational needs.

Essay 2 [25 points]: You have been asked to **design two spacecraft** localization and attitude control and stability system. One shall be a minisatellite in low earth orbit, and the other shall be a large orbiter to be sent to Europa, a moon of Jupiter, for a planetary science mission.

For each spacecraft, at least 350 words per spacecraft, write an essay that must address the following. Any citations used must be cited in line with the text and added to a reference section at the end.

a. **Sensor selection:** For each spacecraft, recommend and justify a suite of sensors to enable the aircraft to sense its state for localization (where am I?) and attitude control. You should also discuss whether the Deep Space Network is needed. *Justify your answer explaining how the components meet the needs for estimating the state (roll, pitch, and yaw), localization, and considering the operating environment(s) and SWAP constraints under which it must operate.*

b. **Actuator Selection:** For each satellite, recommend the actuator(s) necessary to provide spacecraft stability and control. *Justify your answer explaining how the components meet the needs for roll, pitch, and yaw control, and considering the operating environment(s) and SWAP constraints under which it must operate.*

c. **Databus**: recommend a databus type appropriate to the mission type considering the high cost of the orbiter demands a significantly higher bandwidth (more data and control signals between spacecraft systems and payload) and a need for deterministic latency while the minisatellite must have databus with low size weight and power demand. *Justify how the databus is suitable given the constraints of the spacecraft and mission.*

The goal of this problem is to assess your understanding of the components and their limits, the needs for satellite sensing and actuation to enable attitude stability and control (e.g. pointing),

Hint: If you need to see examples of spacecraft to serve as inspiration for spacecraft of similar types and missions, consider the following:

- Orbiter: Mars Reconnaissance Orbiter
- Minisatellite: Satellites that make up NASA's TROPICS CubeSat constellation.

Essay #3 [25 points]: Explain the concept of Gold Codes as they are utilized within the Global Positioning System (GPS), drawing specifically from the provided materials.

- Describe the purpose of PRNs for determining the range between receiver and the satellite-based transmitter.
- Describe the algorithm behind the generation of Gold Codes, including the role of Linear Feedback Shift Registers (LFSRs) and the combination of different generating polynomials.

Multiple Choice

1. **[Multiple choice 2.5 points]:** Skywaves are:

   a. Electromagnetic waves that travel within the ionosphere, therefore, they can reach further distances than the line-of-sight.

   b. Electromagnetic waves that reflect back from the ionosphere, therefore, they can reach further distances than the line-of-sight.

   c. Electromagnetic waves that travel to space, passing through the ionosphere, therefore, they can reach further distances than the line-of-sight.

   d. Electromagnetic waves that reflect back from the clouds, therefore, they can reach further distances than the line-of-sight.

2. **[Multiple choice 2.5 points]** The VOR systems measure:

   a. The bearing of the airplane from a given point (ground station), referenced to True North (geodetic north).

   b. The bearing of the airplane from a given point (ground station), referenced to Magnetic North.

   c. The bearing of your airplane from other aircraft.

   d. The distance of the airplane from a given point (ground station).

Short Answer

3. **[Short Answer – 5 points]:** An aircraft is flying at 30000 ft and the DME equipment reports 5 nmi. How far horizontally is the aircraft from the DME station in nmi? (1 nmi = 6076.12 ft).

4. **[Short Answer – 5 points]:** Explain what radio multipath is, and why it is a problem for avionics. Provide an example.

5. **[Short Answer – 5 points]** If an aircraft is flying to a VOR station heading 270 degrees along the VOR radial 270. The VOR indicator shows the TO flag and the course deviation indicator (CDI) centered.
    - If the pilot turned the aircraft such that the aircraft returns to the 270 radial, but flying in the exact opposite direction, what would the TO/FROM indicator show?
    - If the CDI is three ticks to the right of center, what direction should the pilot turn and by how many degrees?

6. **[Short Answer – 5 points]** Differentiate between standards, regulations, orders, technical standard orders, airworthiness directives, and advisory circulars. What is each's role in aviation safety for operations, technologies, and systems?